

AUTONOMOUS ROBOTIC SYSTEM FOR PIPELINE INSPECTION

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Introduction

- Our project develops an AI-powered autonomous robotic system designed to effectively navigate pipelines for inspection, monitoring, and operational reliability.
- The robotic system intelligently detects cracks and blockages within pipelines, enabling real-time assessments that ensure accurate identification of potential infrastructure issues.

Literature Review

Author	Approach	Type	Description
N. Ananthanarayanan, V. S. S. Kumar, N. R. Raajan, et al	Convolutional Neural Network (SSD-Inception v3) with a reconfigurable teleoperated robot for drain inspection using transfer learning.	Research Article	A teleoperated, reconfigurable robot uses (Raptor) SSD-Inception v3 CNN for detecting drain-blocking objects (bushes, trash, silt) via transfer learning. Achieves 91.4 percentage accuracy at 18 FPS in real-time. Field trials in varied drains demonstrated stable maneuverability, precise mapping, and localization.

Table 1: Literature Review

Literature Review

Author	Approach	Type	Description
Ruwan Ten-nakoon, Reza Ho-seinnezhad, Huu Tran, Alireza Bab-Hadiashar	Deep Convo-lutional Neural Networks with transfer learning and data augmen-tation	Research Paper	Proposes an automated vision-based method using CNNs (ResNet-50 and a shal-low model) for detecting five pipe defect types from CCTV footage, significantly improv-ing inspection accuracy and efficiency.

Table 2: Literature Review

Literature Review

Author	Approach	Type	Description
Saber Kazemi-nasab, Neda Sadeghi, Vahid Janfaza, Moein Razavi, Samira Ziyadidegan, and M. Katherine Banks	localization, mapping, navigation, and inspection methods	IEEE Research Article	This paper reviews state-of-the-art methods for autonomous in-pipe robots focusing on localization, mapping, navigation, and inspection. It covers various mechanical designs, sensors, image processing techniques, and control strategies such as SLAM and Kalman filters, offering a comprehensive guide for designing intelligent, adaptable robots for pipeline inspection and maintenance.

Table 3: Literature Review

Literature Review

Author	Approach	Type	Description
Yang et al., Halfawy et al., Hawari et al., Kumar et al., Li et al., Xie et al., Cheng , Wang, Yin et al., Tan et al. [?]	Traditional CV, CNNs, Faster R-CNN, YOLO variants, feature fusion techniques.	Research article	This paper proposes an improved YOLOv4-based deep learning method for automatic detection of sewer pipe defects. It enhances feature fusion using Spatial Pyramid Pooling and evaluates different loss functions. A custom-labeled dataset is used, achieving high accuracy (mAP 92.3 percentage) in detecting four common sewer defects with improved efficiency.

Table 4: Literature Review

Literature Review

Author	Approach	Type	Description
Niannian Wang, Yihang Duan, Songtao Li, Fan Huang, Lei Qiao.	An advanced YOLOv7-DeepSort (YOLO-DS) based detection and tracking system with a fusion module combining detection and ReID features	IEEE Research Article	The study proposes an automated system for detecting and tracking drainage pipeline defects using YOLOv7 and DeepSort. Achieving 91.1 percentage mAP and 172 FPS, it ensures robust detection under lens rotation, jitter, and blur, and integrates an information management platform for defect reporting and maintenance planning.

Gaps identified

1. Manual pipe inspection is dangerous and unsafe for humans.
2. Existing methods do not provide automatic crack detection.
3. Real-time crack detection and alert systems are not available.
4. Traditional systems lack AI-based intelligent crack detection.

Problem Statement

Manual inspection of pipelines are inefficient and hazardous, creating the need for an autonomous, AI-powered robotic system to detect and monitor clogged and poorly maintained pipeline systems in real time.

Objectives

- Develop an AI-powered autonomous robot to navigate and inspect pipelines.
- Enable real-time monitoring and reporting.

- **Phase 1: Dataset Collection and Preparation**

1. Collect crack and no-crack images from dataset (Concrete Crack Dataset).
2. Label and preprocess images for training the CNN model.

- **Phase 2: AI Model Development**

1. Design and build a CNN model for crack detection.
2. Train, validate, and convert the model to TensorFlow Lite for Raspberry Pi.

- **Phase 3: Hardware and System Integration**

1. Assemble robot with Raspberry Pi, camera, motors, and sensors.
2. Deploy trained CNN model on Raspberry Pi.
3. Configure ESP32 for wireless communication and alert system.

- **Phase 4: Testing and Evaluation**

1. Test robot movement and crack detection inside pipe setup.
2. Evaluate detection accuracy and system performance.
3. Optimize speed, detection threshold, and system stability

Proposed System

- **Dataset** : A crack image dataset is collected from publicly available sources and used to train the model. The dataset includes crack and no-crack images for accurate detection.
- **Model building** :A lightweight CNN model is designed and trained using TensorFlow to detect cracks. The trained model is converted to TensorFlow Lite and deployed on Raspberry Pi for real-time crack detection.
- **Evaluation** : The system is tested in a pipe setup to evaluate crack detection accuracy, real-time performance, and overall robot inspection efficiency.

Architecture of Proposed System

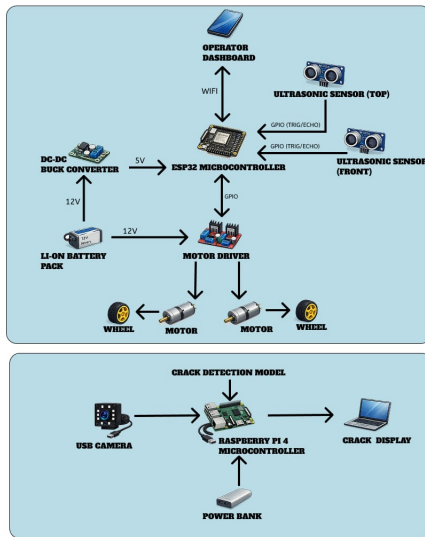


Figure 1: Architecture of proposed method

Architecture of Proposed System

- **Camera Input** : A USB webcam captures real-time images of the pipe interior. The captured frames are sent to the Raspberry Pi for processing and crack detection.
- **Raspberry Pi 4 (Central Processing Unit)** : Raspberry Pi acts as the main controller of the system. It receives camera input, runs the CNN model using TensorFlow Lite, and detects cracks. It also controls motors and processes sensor data.
- **CNN-Based Crack Detection Module** : The CNN model performs real-time crack detection on captured images. If a crack is detected, the system generates an alert and sends the information to the operator

Architecture of Proposed System

- **Ultrasonic Sensors (Obstacle Detection)** : Ultrasonic sensors detect obstacles in front and rear of the robot. This data is sent to Raspberry Pi to avoid collisions.
- **Motor Driver and Drive System** : The L293D motor driver receives control signals from Raspberry Pi. It controls the DC motors for robot movement inside the pipe.
- **ESP32 Communication Module** : ESP32 provides wireless communication between robot and operator. It sends crack alerts and live status to the operator dashboard via Wi-Fi.

Architecture of Proposed System

- **Operator Dashboard (Monitoring System)** : The operator receives crack alerts and robot status on laptop or mobile. This enables real-time monitoring of the pipe inspection process.
- **Power Supply System** : A 12V Li-Ion battery provides power to motors and electronics. A DC-DC buck converter regulates voltage for Raspberry Pi and ESP32.

Hardware Requirements

- **Raspberry Pi 4 (4GB)** – Main processing unit for CNN model and system control
- **USB Webcam (1080p)** – Captures real-time images for crack detection
- **DC Gear Motors (12V) with Robot Chassis** – Provides robot movement inside pipes
- **Motor Driver (L293D)** – Controls speed and direction of motors
- **ESP32 Microcontroller** – Enables wireless communication and alert transmission
- **Ultrasonic Sensors (HC-SR04)** – Detect obstacles and avoid collisions
- **Li-ion Battery Pack (12V, 5000mAh)** – Powers the robot and electronic components
- **DC-DC Buck Converter (LM2596)** – Regulates voltage for Raspberry Pi and ESP32
- **LED Ring Light** – Provides illumination inside dark pipes

Software Requirements

- **Operating System:** Raspberry Pi OS
- **Programming Language:** Python
- **Libraries and Frameworks:**
 - OpenCV – Image capture and processing
 - TensorFlow Lite – CNN model deployment
 - NumPy – Image and data processing
 - RPi.GPIO – Motor and sensor control
 - PySerial – Communication between Raspberry Pi and ESP32
 - Flask / SocketIO – Real-time monitoring dashboard

- This dataset contains two classes: crack and no-crack. These categories represent structural surface conditions and help in identifying damaged and undamaged areas.
- The dataset includes 40,000 images, with 20,000 crack images and 20,000 no-crack images. It enables accurate training of the CNN model for automatic crack detection in pipeline inspection systems.
- <https://www.kaggle.com/datasets/arunrk7/surface-crack-detection>

Dataset Sample



Figure 2: dataset images

Model Flow Diagram

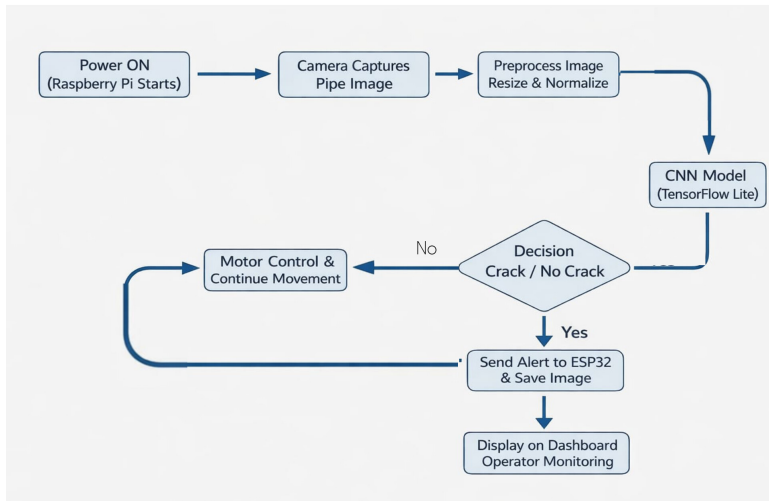


Figure 3: Model Work Flow

Model Flow Diagram

- Image Acquisition and Processing: A USB webcam captures real-time images inside the pipe, which are sent to the Raspberry Pi 4 for preprocessing (resizing to 224×224 and normalization).
- Crack Detection Using CNN: The preprocessed images are fed into a TensorFlow Lite CNN model to classify whether a crack is present or not.
- Alert and Communication: If a crack is detected, the decision module generates an alert, and the ESP32 module transmits the detection result and system status to the operator dashboard via Wi-Fi.
- Navigation and Obstacle Avoidance: The motor control system (L293D with motors) drives the robot inside the pipe, while ultrasonic sensors detect obstacles and send data to the Raspberry Pi for safe movement

Model Architecture

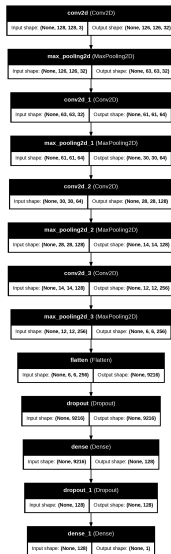


Figure 4: CNN Architecture

CNN Model Architecture

- The model uses a custom Convolutional Neural Network (CNN) for binary crack detection optimized for edge devices.
- Input images are resized to $128 \times 128 \times 3$ to balance accuracy and computational efficiency.
- The architecture consists of four convolutional blocks with increasing filters (32, 64, 128, 256) and MaxPooling layers.
- High-level features are flattened into a vector and passed through dense layers for classification.
- Dropout layers (0.5 and 0.3) are used to reduce overfitting and improve generalization.
- The final layer uses sigmoid activation to output crack probability.
- The dataset is split into 70% training, 20% validation, and 10% testing.
- Data augmentation techniques such as rotation, zoom, flip, and shear improve robustness.
- The model is trained using Adam optimizer with binary cross-entropy loss and achieves 99% accuracy.

Motion Control and Obstacle Detection

- Motion control and obstacle detection is controlled using an ESP32 microcontroller.
- The ESP32 acts as a web server and creates a WiFi access point for user interaction.
- A web-based control panel allows switching between auto mode and manual mode.
- PWM signals from ESP32 regulate the speed of the motors.
- The L293D motor driver is used to control two DC motors for robot movement.

Manual Mode

- In manual mode, the user controls robot movement using buttons like forward, backward, left, right, and stop.
- Each button press sends an HTTP request to the ESP32, which triggers corresponding motor actions.

Auto Mode

- In auto mode, ultrasonic sensors are used for environment sensing.
- The top sensor checks if the robot is inside the pipe by comparing with the set pipe diameter.
- The front sensor detects obstacles, making the robot move backward if an object is too close, otherwise move forward.

Model Performance Evaluation

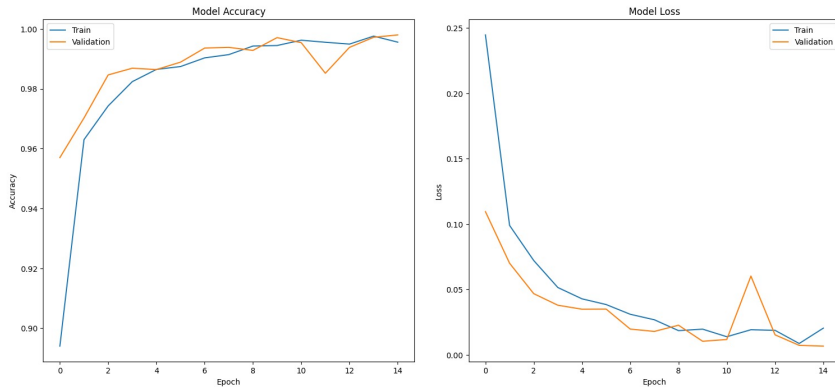


Figure 5: accuracy and loss graph

Model Performance Evaluation

- The evaluation graph shows training and validation accuracy and loss across epochs.
- Training accuracy increases steadily and reaches 99.06%.
- Validation accuracy stabilizes around 99.04%, indicating consistent performance.
- The close match between training and validation accuracy shows good generalization.
- Training loss decreases from 0.25 to 0.04 over time.
- Validation loss reduces from 0.12 to approximately 0.03.
- The decreasing loss values indicate proper model convergence.
- The small gap between training and validation results suggests minimal overfitting.
- Overall, the CNN model achieves stable learning and reliable crack detection performance.



Figure 6: robot

Result

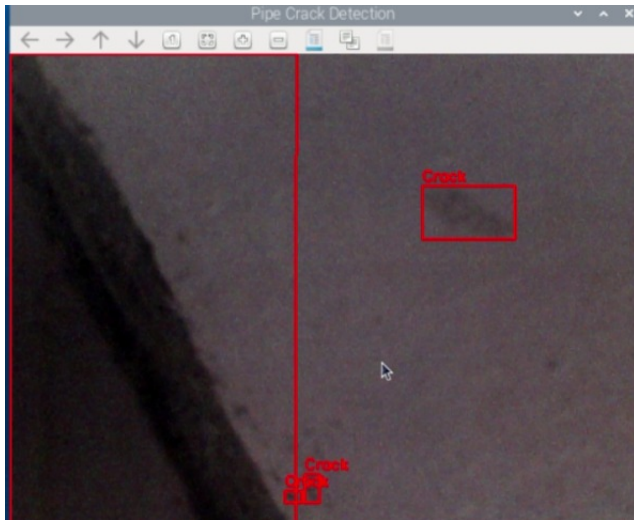


Figure 7: detection

User Interface

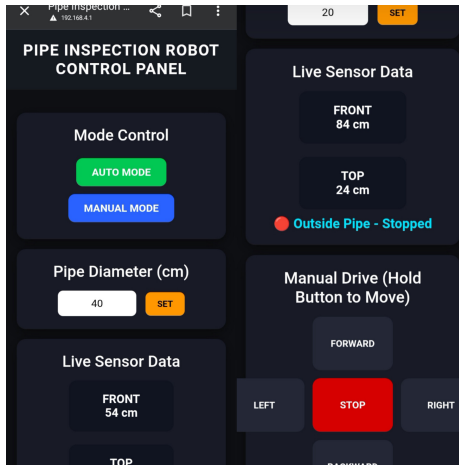


Figure 8: ui

User Interface

- The user interface is a web-based control panel hosted on the ESP32 and accessed via a browser.
- The Mode Control section provides buttons to switch between Auto Mode and Manual Mode.
- The Pipe Diameter input field allows the user to set the required pipe size for auto operation.
- A Live Sensor Data section shows real-time distance readings from front and top ultrasonic sensors.
- A status indicator displays whether the robot is inside or outside the pipe and its current state.
- The Manual Drive section includes directional buttons for forward, backward, left, right, and stop.
- Each button sends a command to the ESP32 through HTTP requests for real-time control.

Budget Overview

Component	Qty	Cost (INR)	Purpose
Raspberry Pi 4 (4GB)	1	₹7,500	Main processor
ESP32 Dev Board	1	₹450	Wi-Fi communication
USB Webcam 1080p	1	₹900	Video capture
12V DC Gear Motors	4	₹1,200	Robot locomotion
L293D Motor Driver IC	2	₹100	Motor control
HC-SR04 Ultrasonic Sensor	2	₹160	Obstacle detection
Li-Ion Battery (12V, 5000mAh)	1	₹1,200	Power supply
DC-DC Buck Converter (LM2596)	1	₹120	Voltage regulation
LED Ring Light	1	₹300	Pipe illumination
Robot Chassis (3D Printed)	1	₹800	Physical frame
Wheels, Wires, PCB, Connectors	—	₹900	Assembly parts
TOTAL		₹13,630	

Figure 9: cost breakdown

Conclusion

The AI-based pipe inspection robot successfully detects cracks in real time using a CNN model and Raspberry Pi. It improves inspection safety, accuracy, efficiency, and enables early maintenance of pipeline infrastructure.

- **NVIDIA Jetson Nano:** Enables faster YOLOv8 inference and real-time crack localization with bounding boxes.
- **GPS Module (NEO-6M):** Provides geographic coordinates of crack locations in outdoor pipelines
- **Raspberry Pi Camera v3 (Wide) :** 120° FOV for better pipe coverage
- **IMU Module MPU-6050 :** Tilt compensation and navigation logging
- **Li-Ion Battery with USB-C charging BMS :** Safer and more convenient charging

References

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THANK YOU